

Total No. of Questions : 6]

SEAT No. :

P5452

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BE/Insem./Oct.-23

B.E. (Mechanical) (Semester - I)

OPERATIONS RESEARCH

(2012 Pattern) (Elective - II)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4 and Q.5 or Q.6.
- 2) Answer THREE questions.
- 3) Use of non-programmable scientific calculator is allowed.
- 4) Assume suitable data where ever necessary.
- 5) Figures to the right indicate full marks.

Q1) A gear manufacturing company received an order for three specific types of gears for regular supply. The management is considering to devote the available excess capacity to one or more of the three types, say, A, B and C. The available capacity on the machines which might limit output and the number of machine hours required for each unit of the respective gear is also given below : [10]

Machine Type	Available machine hours/week	Productivity in machine hours/unit		
		Gear A	Gear B	Gear C
Gear Hobbing m/c	250	8	2	3
Gear Shaping m/c	150	4	3	0
Gear Grinding m/c	50	2	-	1

The unit profit would be Rs. 20, Rs. 6 and Rs. 8 respectively for gear A, B and C. Find how much of each gear the company should produce in order to maximize profit?

OR

Q2) a) A TV dealer finds that the cost of a TV in stock for a week is Rs. 30 and the cost of a unit shortage is Rs. 70. The probability distribution of weekly sales of a TV model is as follows : [6]

Weekly Sales	0	1	2	3	4	5	6
Probability	0.10	0.10	0.20	0.25	0.15	0.15	0.05

From this data, calculate Expected Value with Perfect Information (EVPI)

P.T.O.

- b) Explain the term 'Decision Tree' and discuss on general approach, used in decision tree analysis. What are the advantages and limitations of the decision tree approach? [4]

Q3) A cement factory manager is considering the best way to transport cement from his three manufacturing centres P, Q, R to the depots A, B, C, D and E. Based on the weekly production and demands along with transportation costs per ton as given in below table, what should be the distribution program? Use Vogel's Approximation Method for initial basic feasible solution and UV method for optimisation [10]

	A	B	C	D	E	Tons
P	4	1	3	4	4	60
Q	2	3	2	2	3	35
R	3	5	2	4	4	40
	22	45	20	18	30	

OR

Q4) a) Company has four territories namely I, II, III and IV open and available for assignment to the four salesmen namely A, B, C, D. The anticipated annual sales figures are as per below table. Determine optimum assignments using Hungarian method to achieve maximum total sale. [6]

	I	II	III	IV
A	42	35	28	21
B	30	25	20	15
C	30	25	20	15
D	24	20	16	12

- b) What is Transshipment model? Explain with schematic diagram the transshipment problem with some transient nodes in between sources and destinations. [4]

Q5) a) Reduce the following game by dominance and find the game value. [6]

		Player B			
		1	2	3	4
Player A	1	3	2	4	0
	2	3	4	2	4
	3	4	2	4	0
	4	0	4	0	8

- b) Explain 'Payback Period' method of Investment analysis [4]

OR

Q6) a) Following figures are related to a manufacturing company. **[6]**

- i) Variable Cost per Unit = Rs. 8 /-
- ii) Sell Price per unit = Rs. 14 /-
- iii) Total Units sold = 50,000
- iv) Fixed Cost = Rs. 12,000 /-

Based on the above information, calculate -

- I) Profit – Volume (P/V) Ratio
 - II) B.E.P. in sales
 - III) Margin of Safety
- b) Explain terms ‘Pure Strategy’ and ‘Mixed Strategy’ from Game Theory. **[4]**

